

JEE MAIN REVISION MASTER: WAVES SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 96 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Wave Equation & Velocity of Sound	66 PYQs	Wave Motion
Very High	Standing Waves, Resonance & Doppler Effect	40 PYQs	Interference & Acoustics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Wave Equation & Velocity of Sound	$y(x,t) = A \sin(kx \mp \omega t + \phi)$, $v = \frac{\omega}{k} = \lambda$, $v_{\text{string}} = \sqrt{\frac{T}{\mu}}$, $v_{\text{gas}} = \sqrt{\frac{\gamma RT}{M}}$
Standing Waves, Resonance & Doppler Effect	$f_{\text{open}} = \frac{nv}{2L}$; $(n=1,2,3)$, $f_{\text{closed}} = \frac{(2n-1)v}{4L}$; $(n=1,2,3)$, $f' = f \left(\frac{v \pm v_0}{v \mp v_s} \right)$, $f_{\text{beat}} = f_1 - f_2 $

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Wave Equation & Velocity of Sound (Very High Yield — 66 PYQs)

Core Principle: Progressive traveling harmonic waves, wave vector $k = 2\pi/\lambda$, and Laplace-corrected speed of sound in gases.

■ **Caution / JEE Pitfall:** Maximum particle velocity $v_{\text{(particle, max)}} = A\omega$ is completely different from wave propagation velocity $v = f\lambda$.

■ **Speed Trick:** Wave travels in positive x-direction if signs of kx and ωt are OPPOSITE: $\sin(kx - \omega t)$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "The equation of a plane progressive wave is given by $y=5\cos\pi(200t-(x/150))$ where x and y are in cm and t is in second. The $v_{\dots}$ " [Ans: D]
- [2026 Morning Shift] Q#1 (MCQ): "Two strings (A,B) having linear densities $\mu_{\text{(A)}}=2 \times 10^{-4}\text{kg/m}$ and, $\mu_{\text{(B)}}=4 \times 10^{-4}\text{kg/m}$ and lengths $L_{\text{(A)}}=2.5\text{m}$ and ..." [Ans: B]
- [2025 Morning Shift] Q#1 (MCQ): "Two harmonic waves moving in the same direction superimpose to form a wave $x=a\cos(1.5t)\cos(50.5t)$ where t is in seconds. Find the period $w_{\dots}$ " [Ans: N/A]

3.2 Standing Waves, Resonance & Doppler Effect (Very High Yield — 40 PYQs)

Core Principle: Stationary waves in strings and organ pipes, beat frequency, and Doppler frequency shift for moving sources and observers.

■ **Caution / JEE Pitfall:** A closed organ pipe produces ONLY odd harmonics (1, 3, 5...). The first overtone is the 3rd harmonic ($3v/4L$).

■ **Speed Trick:** Doppler rule: When distance decreases, apparent frequency INCREASES; when distance increases, apparent frequency DECREASES.

Representative Source PYQs:

- **[2026 Evening Shift] Q#2 (MCQ):** "The fifth harmonic of a closed organ pipe is found to be in unison with the first harmonic of an open pipe. The ratio of lengths of closed p..." **[Ans: N/A]**
- **[2026 Evening Shift] Q#3 (MCQ):** "A point source is kept at the center of a spherically enclosed detector. If the volume of the detector increased by 8 times, the intensity w..." **[Ans: B]**
- **[2026 Evening Shift] Q#4 (MCQ):** "In an open organ pipe v_3 and v_6 are 3rd and 6th harmonic frequencies, respectively. If $\frac{v_6}{v_3} = \frac{1}{3}$..." **[Ans: B]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)

The equation of a plane progressive wave is given by $y = 5\cos\pi(200t - (x/150))$ where x and y are in cm and t is in second. The velocity of the wave is _____ m/s.

- (A) 120
- (B) 150
- (C) 200
- (D) 300 **[Correct]**

Correct Answer: D

Solution Analysis: A plane progressive wave moving in the positive x -direction can be represented as: $y = A\cos(\omega t - kx)$ Where, ω is the angular frequency, k is the propagation constant (wave number). The given equation is $y = 5\cos\pi(200t - (x/150))$. $y = 5\cos(200\pi t - (\pi x/150))$ By comparing this with $y = A\cos(\omega t - kx)$, $\omega = 200\pi$ rad/s and $k = (\pi/150)$ rad/cm Wave velocity is the distance travelled by the wave per unit time. It is related to angular frequency and wave number by the relation: $v = (\omega/k)$ Substituting the values of ω and k : $v = \frac{200\pi}{\pi/150} = 300$ m/s

Benchmark Problem 2: JEE Main 2026 (Online) 24th January Evening Shift (Q#3)

A point source is kept at the center of a spherically enclosed detector. If the volume of the detector increased by 8 times, the intensity will

- (A) increase by 64 times
- (B) decrease by 4 times **[Correct]**
- (C) decrease by 8 times
- (D) increase by 8 times

Correct Answer: B

Solution Analysis: The volume (V) of a sphere of radius R is given by the formula : $V = \frac{4}{3}\pi R^3$ (i) According to the problem, the volume increases by 8 times ($V' = 8V$) : $8V = \frac{4}{3}\pi (R')^3$... (ii) $\Rightarrow 8 \times \frac{4}{3}\pi R^3 = \frac{4}{3}\pi (R')^3 \Rightarrow R' = 2R$ The intensity (I) of a point source (of power P) is the power spread over the surface area of the sphere. The formula for intensity is : $I = \frac{P}{A} = \frac{P}{4\pi R^2} \Rightarrow I \propto \frac{1}{R^2}$ Using the new radius $R' = 2R$:

$$\frac{I'}{I} = \frac{(1/R'^2)}{(1/R^2)} \Rightarrow \frac{I'}{I} = \frac{1}{4} \Rightarrow I' = \frac{I}{4}$$